Results

The error analysis covers the 30 public law and civil law tasks of the de-core-45 study with eight arms per task (Table 1). Every content criterion that the judge committee rated FAIL was assigned to an error class independently by two AI annotators; an assigned FAIL is called a finding. The tables count findings, not errors: the same omission can be captured by several criteria of one answer. Findings on which the annotators disagree are open until the author decides them and are excluded from the frequencies.

Main findings

- Four out of five findings are omissions: K1 accounts for 78% of the counted findings. Misstated rules (K3, 7%), misapplied facts (K2, 7%) and citation errors (K5, 5%) follow far behind; K4, K6 and K7 together make up 0.8%.
- Within K1, in 59% of the findings with a level the subject matter of the required examination is not named at all (K1a); in 26% the standard is missing, in 14% the application, and only in 1% the conclusion alone.
- The distribution is similar across the eight arms (Table 5). The K3 share ranges from 3% (Sol baseline) to 10% (DeepSeek baseline).
- The two annotators agree on the class of 94% of the findings (α 0.850), with the K1 levels included on 80%. Agreement is lower in public law than in civil law (Table 2).
- 2% of the counted findings are FAILs that the answer text does not confirm (OK), a finding about the automatic rating.

Scope and agreement

Table 1 gives the scope of the analysis, Table 2 the agreement of the two annotators by area of law.

Error distributions

Table 3 gives the class frequencies by area of law and Table 4 the four levels of K1. Table 5 gives the class shares per arm. Arms: SB Sol baseline, SA Sol agent, TB Terra baseline, TA Terra agent, DB DeepSeek baseline, DA DeepSeek agent, CN Codex without web, CW Codex with web.

Table 1: Scope: tasks, answers and content ratings of the eight arms.

Area of law	Tasks	Answers	Content ratings	Content FAILs	FAIL share
Public law	15	119	4,370	1,447	33.1%
Civil law	15	120	4,552	1,997	43.9%
Total	30	239	8,922	3,444	38.6%

Note. Eight arms per task; one arm has no submission for one task. A content rating is one content criterion of one answer rated PASS or FAIL by the judge committee. Every content FAIL is one finding of the annotation.

Table 2: Agreement of the two annotators on the class of each finding.

	n	Class level		With K1 levels	
		Agreement	α	Agreement	α
Public law	1,447	92.1%	0.830	78.0%	0.727
Civil law	1,997	94.8%	0.867	81.3%	0.734
Total	3,444	93.7%	0.850	79.9%	0.735

Note. Percent agreement and Krippendorff's α (nominal). The class level merges K1a to K1d into K1.

Table 3: Class frequencies of the findings by area of law.

Class	Public law	Civil law	Total	Share
K1: examination or explanation missing	1,009	1,520	2,529	78.4%
K2: facts misapplied	113	107	220	6.8%
K3: rule misstated	65	160	225	7.0%
K4: facts altered	2	0	2	0.1%
K5: citation missing or wrong	92	74	166	5.1%
K6: self-contradiction	0	3	3	0.1%
K7: designation missing	14	7	21	0.7%
OK: FAIL not confirmed	38	23	61	1.9%
Total	**1,333**	**1,894**	**3,227**	**100.0%**

Note. Findings on which both annotators assign the same class (3,227 of 3,444); the 217 findings with disagreement are open until the author decides them. OK is not an error class: the FAIL is not confirmed by the answer text. Definitions in error-classes.pdf.

Table 4: The four levels of K1: the first missing element of the required examination.

Level	What is missing first	n	Share
K1a	subject matter not named	1,204	58.6%
K1b	standard missing	529	25.8%
K1c	application missing	296	14.4%
K1d	conclusion missing	24	1.2%
K1 with level		2,053	100.0%

Note. Of the 2,529 K1 findings, 2,053 have a level on which both annotators agree; 476 are open at the level. The levels are separated less reliably than the classes (Table 2).

Table 5: Content FAILs and class shares by arm.

Arm	Content FAILs	Share of the arm's counted findings					
		K1	K2	K3	K5	K4, K6, K7	OK
Sol baseline	368	80.1%	6.6%	2.9%	6.9%	1.2%	2.3%
Sol agent	316	79.1%	5.1%	7.1%	4.4%	1.0%	3.4%
Terra baseline	466	79.0%	5.8%	6.0%	6.7%	0.4%	2.0%
Terra agent	483	75.6%	10.2%	7.1%	5.3%	0.4%	1.3%
DeepSeek baseline	620	79.5%	5.1%	10.2%	3.4%	0.8%	1.0%
DeepSeek agent	527	73.3%	8.4%	9.7%	5.1%	1.1%	2.5%
Codex without web	373	83.7%	4.3%	5.4%	4.6%	0.9%	1.1%
Codex with web	291	78.8%	9.2%	3.7%	5.5%	0.7%	2.2%

Note. Content FAILs: all content FAILs of the arm over its submitted answers (30 per arm, DeepSeek agent 29). Shares refer to the arm's counted findings (unanimous or decided) and add up to 100 per row. No model ranking is derived from these numbers.

Errors hidden behind omissions

The order of checks records an omission before a misstated rule or an altered fact at the same place. Table 6 counts how often such an error stands behind a K1 finding.

Table 6: Misstated rules (K3) and altered facts (K4) hidden behind an omission (K1) at the same place.

Annotator	K1 findings	K3	K4	Both	Hits	Hit rate
Claude	2,617	70	17	1	88	3.4%
Codex	2,554	69	29	3	101	4.0%

Note. The order of checks records only the primary error. For every K1 finding the annotator additionally answered whether a K3 or K4 is present at the same place; the answer does not change the class. Both annotators see a hit in 37 of the 2,529 findings both assigned to K1, at least one of them in 144.